Filippo Mugnai

Software Engineer | Backend & Full-Stack | Python, Java, Microservices | Defence & Aerospace

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PROFILE

Software Engineer with 3+ years of experience in enterprise and mission-critical environments (Defence, Aerospace, Healthcare and Public Administration). At aizoOn I contributed, on an international aerospace programme, to migrating a platform from a monolith to Python/FastAPI microservices supporting Machine Learning pipelines; I currently work on independent software verification and validation in the space domain. Solid Java/Spring Boot background and full-stack experience with Vue.js and Angular.

PROFESSIONAL EXPERIENCE

Software Engineer Consultant — May 2026 – present

aizoOn Technology Consulting — Rome

Client Leonardo (May – August 2026): modernisation of a simulation and optimisation platform for an international aerospace programme, geared towards ML model training.

- Contributed to migrating a distributed monolith (~28,000 lines of Python) to FastAPI microservices on PostgreSQL with Keycloak authentication; legacy code analysis and code reviews.
- Designed and built from scratch a microservice for neural-network hyperparameter tuning, released to production.
- Integrated the Vue.js/TypeScript portal end-to-end with the new backend, solving authentication and frontend–backend communication issues.
- Found and fixed a security flaw in authentication, then verified it across the team's other services.
- Since September 2026: independent software verification and validation (ISVV) for a space programme, reviewing code, requirements and tests.

Stack: Python, FastAPI, Pydantic, SQLAlchemy, PostgreSQL, Vue.js, TypeScript, Keycloak (OIDC), Docker, Git/GitLab, GitLab CI, automated testing — Kubernetes/Kubeflow environment

Almaviva — Rome — April 2023 – March 2026

Digital Content & Software Specialist — Italian Ministry of Defence — May 2025 – March 2026

- Development and customisation of CMS components (HTML, CSS, .NET) for digital content and information processes; technical coordination with IT teams and validation of application changes.

Software Developer — Public healthcare and payments — October 2024 – May 2025

- Functional testing, bug validation and technical documentation for healthcare applications (AGENAS) and PagoPA-related systems; Oracle database queries, UI design in Figma.

Java Developer — Defence & Security Division — April 2023 – September 2024

- Java (JavaFX/Swing) applications for the design and simulation of naval scenarios.
- Backend development in Java/Spring Boot with Hibernate on Oracle/MySQL; Agile teamwork with GitHub and YouTrack.

Stack: Java, Spring Boot, Hibernate, JavaFX/Swing, Oracle SQL, MySQL, Angular, .NET, Figma, GitHub, YouTrack

PERSONAL PROJECTS

Desktop app for personal trainers — 2026

- Application in real use for managing athletes, an exercise catalogue with a muscle map and monthly training plans.
- Encrypted local SQLite storage, repository-based architecture decoupled from the framework; CI pipeline with TypeScript checks, unit and end-to-end tests and secret scanning; AI-assisted development with Claude Code.

Stack: Electron, React, TypeScript, SQLite (encrypted), GitHub Actions

TECHNICAL SKILLS

Languages	Python, Java, TypeScript/JavaScript, C#, C/C++, SQL
Backend	FastAPI, Pydantic, SQLAlchemy, Spring Boot, Hibernate, REST APIs, SSE, microservices
Frontend	Vue.js, React, Angular, Electron, HTML, CSS, Figma
Databases	PostgreSQL, Oracle SQL, MySQL, SQLite
DevOps & Tools	Docker, Git, GitHub, GitLab, GitLab CI, GitHub Actions, Keycloak (OIDC), YouTrack, VS Code, IntelliJ IDEA
Quality & method	Automated testing, code review, Mypy strict, Ruff, Agile/Scrum
Spoken	Italian (native), English (B2)

EDUCATION

BSc in Computer Engineering (in progress) — October 2024 – present

eCampus University — Automation and Artificial Intelligence track

Regional Analyst Programmer Course (500 hours) — September 2022 – March 2023

Istituto Cefi — accredited by the Lazio Region (DGRL G00157) — certified modules: Java/Hibernate/Spring, C/C++, C#, Visual Basic .NET, Programming Techniques

Scientific High School Diploma — September 2016 – July 2021

Liceo Scientifico Poliziano, Rome

CERTIFICATIONS

Machine Learning Specialization – DeepLearning.AI / Stanford University (Coursera), January 2026 · Registered Scrum Basics™ (Scrum Inc., 2024) · Agile/Scrum Fundamentals (Scrum Inc.)

I authorise the processing of the personal data contained in this CV in accordance with Regulation (EU) 2016/679 (GDPR).